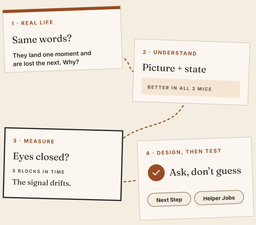
An ASDRP neuroscience portfolio · Amishi Mehta

From neurons to next steps

It started from Amishi's own experience: the same instruction can land one moment and be lost the next. **Is that the brain itself?** Two studies with public brain recordings and two apps later, here is what we found, in three sentences.

What we found → | The whole story



What we found, in three sentences

Each sentence is one finding, and each links to the evidence behind it.

1 · INSIDE A BRAIN

What a brain is doing changes how it responds to the same thing.

In three mice, knowing whether the mouse was alert and moving made its brain cells **25–33% easier to predict**, for the very same pictures.

The mouse study →

2 · FROM OUTSIDE THE HEAD

But that state is hard to read from outside.

A model reading one person's brain waves could tell eyes open from closed on one stretch, then drifted to **a coin flip within two minutes**. Brain-wave bands did not fix it.

The EEG study →

3 · IN PRACTICE

So tools should ask, not guess, and keep each step small.

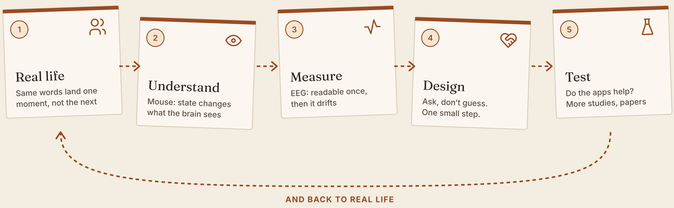
Next Step asks **"How much energy do you have today?"** and on a low day shows just the next step. Helper Jobs gives one picture step at a time.

The design →

What comes next: test whether that actually helps the people the apps are for, and take the science further with an ASDRP mentor. The plan →

One loop, five chapters

It starts in real life and ends back there. Each chapter answers the question the last one raised.



1 · REAL LIFE

Same words, different moment.

It started from Amishi's own experience: a step that is easy when you are calm can be lost when you are tired, rushed or overloaded. Is that the brain itself?

To find out: look inside a brain while its state changes.

2 · UNDERSTAND

A mouse's state changes how its brain sees.

Mice watched pictures while thousands of brain cells were recorded. We guessed what each cell would do from the picture alone, then from the picture plus how alert and active the mouse was.

Found: knowing the mouse's state made our guesses 25–33% better, in all three mice. Read chapter 2 →

3 · MEASURE

Can sensors read state in a person?

A cap of 14 sensors on one person's head for two minutes (EEG). Closing your eyes is one of the clearest changes it picks up. Can a simple model tell when?

Found: on one stretch, often (0.625, where 0.5 is a coin flip). Across the whole recording it drifted to 0.478. Brain-wave bands did better (0.566) but still dipped below guessing. Read chapter 3 →

4 · DESIGN

From findings to features.

If state matters but cannot be read from outside, an app should ask, not guess. It should show less on a low-energy day, and make pausing and coming back easy.

Built into: Next Step, by Amishi Mehta, for schoolwork, and Helper Jobs, a family project, built for real jobs at Pragnya's Sunday outings. Read chapter 4 →

5 · TEST, THEN GO FURTHER

Where this goes next.

One follow-up done, three more data experiments, design studies with Pragnya volunteers and families, and papers others can check.

The aim: one research direction, where neuroscience and design meet. Read chapter 5 →

In short: state matters, it is hard to read from outside, so ask and keep steps small.
Three mice and one person are a start, not a rule for everyone.

What this does not show yet: whether the pattern holds in more animals and more people, what causes it, or whether the apps help. Each of those is an experiment in chapter 5.

Who did what

The studies were done by Amishi with her father, who helped shape the questions, review the code and fix problems. Software tools, including AI assistants, helped build this website and the apps. The contribution ledger lists each part.

What it is not

It is not a medical study and it does not diagnose anyone. Three mice and one person cannot speak for everyone, and being able to predict something is not the same as knowing what causes it.

CHAPTER 2 · UNDERSTAND · MOUSE STUDY

Does a mouse's state change how its brain responds to what it sees?

Mice watched pictures while scientists recorded thousands of brain cells in the part of the brain that handles seeing. We tried to guess what each cell would do, first from the picture alone, then from the picture plus two signs of the mouse's state: how wide its pupils were and how fast it was running. If state did not matter, the extra signs would not help.

It mattered, in all three mice. Knowing the mouse's state made our guesses a quarter to a third better (25–33%), on pictures the model had never seen.
This shows state and seeing are linked. It does not yet show which causes which.

The results

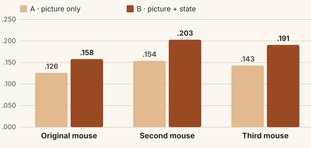


HOW TO READ THE SCORE

How closely the guess follows the real brain cell.

0 means the guess and the cell have nothing in common; 1 would be a perfect match. Brain cells are noisy, so even good guesses score low. What matters is the jump from A to B.

The score is a correlation, averaged over every recorded brain cell.

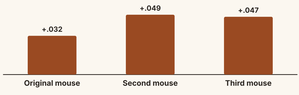


RESULT

Every mouse is easier to predict once we know its state.

The picture explains part of what the seeing part of the brain does. Whether the mouse is alert and moving explains more, in every animal.

Tested on pictures the guessers never saw.



HOW MUCH STATE ADDS

Every mouse gains, by different amounts.

Pupil size helps all three mice. How much running adds changes from mouse to mouse, so each mouse is shown on its own.

Gain = guesser B's score minus guesser A's score.

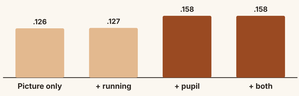
Inside the first mouse

From Amishi's saved results and report in Amishi123/neuroscience-project, redrawn here. These are for the first mouse only.

90%
of brain cells were guessed better once state was added

+0.017
typical gain per cell (the middle value)

+0.35
the biggest gain for a single cell

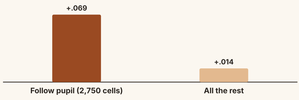


WHICH PART OF STATE?

It is the pupil, not the running.

Adding running barely moved the score. Adding pupil size gave almost the whole gain. This mouse ran on only about 5% of trials, so running had little to say.

Score on unseen pictures, first mouse.

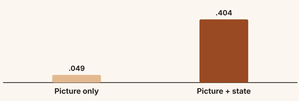


WHICH CELLS GAIN?

The cells that follow alertness gain most.

Cells whose activity rises or falls with pupil size gained about five times as much as the rest.

A cell "follows pupil" when its activity and pupil size move together (correlation above 0.1), as in Amishi's report.

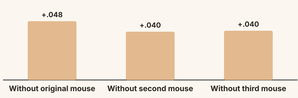


ONE CELL

From noise to a pattern.

Brain cell 3035 was almost impossible to predict from the picture alone. Knowing the mouse's state raised its score from 0.05 to 0.40.

Score for this one cell on unseen pictures.



LEAVE ONE MOUSE OUT

The gain stays without any one mouse.

So the pattern does not hang on one recording. It is a check, not a promise about all mice.

Average gain over the two mice that are left.

Code and data
The analysis is in Amishi123/neuroscience-project (her notebooks 01–05). The numbers on this page are copied from its saved results, and an automatic check confirms they still match. Data: Willeke et al. (2022), *The Sensorium competition on predicting large-scale mouse primary visual cortex activity*, arXiv:2206.08666.

What it does not show
Three mice cannot speak for all mice. A better guess does not show that being alert or running causes the brain to change. Other things that happen at the same time could explain it.

What chapter 2 tells us: the same picture lands differently depending on the state the mouse is in. So the next question is whether that state can be read from outside the head, in a person. Chapter 3 →

Can sensors on the head tell when someone's eyes are closed?

EEG is a cap of small sensors that pick up the brain's electrical activity through the scalp. Closing your eyes is one of the clearest changes it can see. If a simple model cannot read even that reliably for two minutes, quieter states like tiredness will be harder still.

On one stretch, yes. Over time, no. On the last part of the recording the model scored 0.625, and 0.728 once four broken readings were capped. But tested on five stretches across the recording, it averaged 0.478: about a coin flip.
Letting it see the last few moments as well did not help.

The results

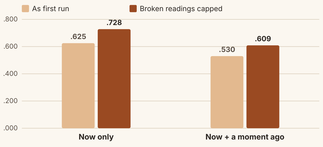


HOW TO READ THE SCORE

Where the results sit between guessing and always right.

Anything near 0.5 is no better than flipping a coin. Below 0.5 means the model is wrong more often than right, so the pattern it learned has flipped.

The score is called AUC.

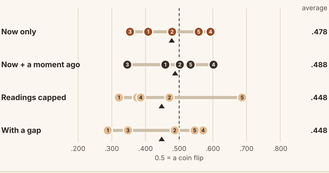


THE LAST STRETCH

Good on one stretch, better once broken readings are capped.

Four readings were broken: up to 715,897 when normal is about 4,000. Capping each sensor at the range seen while learning lifts the score from 0.625 to 0.728.

We added this check after the first results, so the new numbers sit next to the old ones, not in place of them.

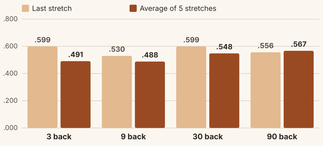


FIVE STRETCHES IN TIME

Across the recording, it drifts to a coin flip.

Scores land on both sides of 0.5, and some fall well below it. Being worse than guessing means the pattern the model learned earlier had flipped by later on.

So the signal drifts within two minutes. Capping broken readings and leaving a gap do not fix it.



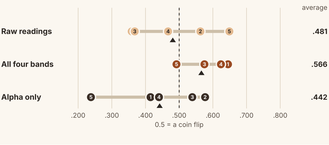
HOW FAR BACK TO LOOK

Looking further back seems to help, but we did not pick a winner.

We chose to try 3, 9, 30 and 90 readings back before seeing any results. Ninety does best on average, but picking it now would mean tuning to this one recording.

Next: compare these on new recordings before choosing.

Better on average, but it still did not hold up. All four bands averaged **0.566** across the five stretches, against **0.481** for the raw readings. But two stretches still dipped just below a coin flip (0.492 and 0.493), so by the rule we set beforehand it did not hold up. Alpha alone did worse (0.442).
Mixing up the answers gave 0.500, as it should.



EXPERIMENT 2

Brain-wave bands help, but not enough to rely on.

Bands lift the average and the worst stretch, but a model that is sometimes worse than guessing is still not something to build an app on. This is why the apps ask the person.

With a one-second gap between learning and testing. Without the gap: 0.571 for bands, 0.529 for raw readings. Plan written before running: EXPERIMENT-2-PLAN.md in the source code.

Rerun it

`python3 projects/eeg-eye-state/src/analyze.py` writes `results/metrics.json`, `src/analyze_bands.py` writes `results/bands.json` for experiment 2, and `python3 website/build_results.py` turns it into these charts. See the full report. Data: Roesler, O. (2013). *EEG Eye State*. UCI Machine Learning Repository, doi:10.24432/C57G7J.

What it does not show

One person and one short recording cannot show how this works for other people. The only thing measured is eyes open or closed, not attention, tiredness or any condition.

What chapter 3 tells us: a sensor that is right on one stretch and wrong on the next, even using brain-wave bands, is not something to build a tool on. So the apps ask the person instead of guessing. Chapter 4 →

CHAPTER 4 · DESIGN · HUMAN–COMPUTER INTERACTION

From findings to features.

Chapter 2 showed that a brain's state changes what gets through. Chapter 3 showed that state is hard to read from outside. Each finding became a design rule, and each rule is a real feature in two working apps.

Four findings, four design rules

Read each card top to bottom: what we found, the rule it gives, and where the rule lives.

FROM CHAPTER 2

Show less when energy is low.

Found: the same picture lands differently depending on the brain's state.

In the apps: Next Step asks "How much energy do you have today?" On "A little", it shows just the next step. Helper Jobs shows one picture step at a time, read aloud.

FROM CHAPTER 3

Ask, don't guess.

Found: a sensor read eye state on one stretch, then drifted to a coin flip. Brain-wave bands did not fix it.

In the apps: the person tells the app how they are doing. The energy check is optional, Help is a button they press, and nothing happens until they tap.

FROM CHAPTER 3

Expect change halfway through.

Found: within two minutes, the link between signal and state shifted.

In the apps: Next Step has Save & pause, a note of where you stopped, and "Pick up where you left off". Helper Jobs reminds helpers to take breaks.

FROM REAL LIFE

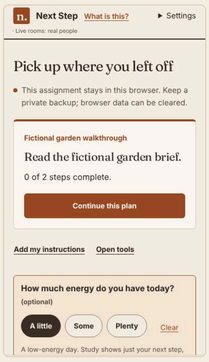
When stuck, make it smaller.

Seen: a task that is too big all at once is where people stop.

In the apps: Next Step's "I'm stuck" offers to make the step smaller, ask a question or take a break. Nothing is sent, and your place is kept.

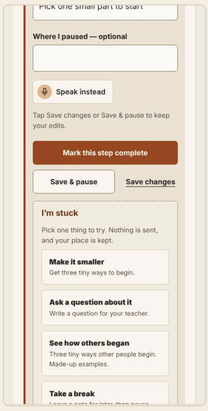
The rules, on real screens

Screens from the working apps, with made-up examples.



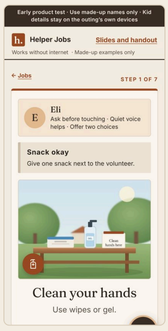
Show less when energy is low

The person picks their energy; the plan shrinks to one step.



Pause, and make it smaller

Save & pause keeps your place; "I'm stuck" shrinks the step.



One picture step, read aloud

A few words, a photo of the real place, and a Help button.

Two apps, built for real life

Both follow the same rules: one step on the screen, big type, a picture, read aloud, and only made-up people in the examples. Swipe to see both.

BY AMISHI MEHTA · FOR SCHOOLWORK

Next Step

For young people who find long instructions, interruptions or explaining their needs hard. Made with neurodivergent students in mind.

What it does: turns a long assignment into one next step, practises words for asking for help, and keeps a note of where to restart. Open Next Step → Its slides and handout →

FAMILY PROJECT · FOR PRAGNYA OUTINGS

Helper Jobs

Built to give neurodivergent adults a real job at kids' Sunday outings, with a volunteer always close by.

What it does: picture steps read aloud in English, Hindi or Spanish, break reminders, and Help that reaches the lead's phone in about a second. Open Helper Jobs → Its slides and handout →

The loop on a phone

The phone tour tells the same five chapters, one screen at a time, in the same design as the apps.

Careful claims

The studies suggest these rules; they do not prove the apps work. These are early test versions with made-up people in every example, not tested in a formal study, and they do not diagnose or treat anything.

Closing the loop: do these designs actually help the people they are for? That is where the research goes next. Chapter 5 →

CHAPTER 5 · TEST, THEN GO FURTHER

Where this goes next.

The loop does not stop at two studies and two apps. Each turn adds a study, a better design and a write-up others can check. Anything not yet done is labelled as a plan.

The road ahead

Each step builds on the last: data first, then people, then the two together.

NOW · DONE

Two studies, one follow-up, two apps.

The mouse and EEG studies, every result checked and rerunnable, plus experiment 2 on brain-wave bands. Next Step and Helper Jobs working, with the four design rules built in.

Shown: on this site, in the slides and in the phone tour.

NEXT · WITH AN ASDRP MENTOR

Run the three remaining data experiments.

Why the EEG drifts (experiment 4), using the data already here. Then timing (experiment 1, code already written) and groups of brain cells (experiment 3) in the full mouse recordings.

Aim: a first paper, on why the EEG drifts and what brain-wave bands can and cannot fix, for a student research journal such as the *Journal of Emerging Investigators*, plus a poster.

THEN · WITH PEOPLE

Study the design with the people it is for.

Co-design sessions with Pragnya volunteers and families, then a usability test of the four design rules. Consent first, and tasks written down before anyone takes part.

Aim: a second paper, on the design rules, and better apps.

LONGER TERM · THE JOINED-UP QUESTION

Does designing for state help people learn?

Compare the energy-aware design with a fixed one, for the same person over several weeks. Then ask whether the brain findings predict who it helps most.

Aim: one research direction, where neuroscience and human–computer interaction meet.

The rule we keep: a plan stays labelled as a plan until it has code, saved results, a rerun record, its limits written down and a page like the two studies. No paper, poster or result is listed as done before it is.